Differential Equations — Reference

how to read this note

Every method is a chain of steps. Each step is a template with empty colored slots.

Do	How
1. find the pieces	Each lesson opens with a PIECES table. It says where each piece is in your equation, and its color.
2. fill the slots	A piece goes in the slot of the same color: p goes in $\boxed{p}$.
3. one box at a time	Each box is one move. Do not merge two boxes.

Colors in lessons 1, 3, 5 and Bernoulli: $p(x)$ $q(x)$ μ (mu) x_0, y_0 $a(x)$ or n. Lessons 2, 4 and the homogeneous method set their own colors in their PIECES tables.

the master idea

Rearrange until each side is integrable.

Method	Rearrange it until
linear (L3)	the left side is one derivative, $(\mu y)'$
separable (L4)	all y is with dy, all x is with dx
Bernoulli (L6)	it is linear in $v = y^{1-n}$. Then L3
homogeneous (L6)	it is separable in $v = y/x$. Then L4
exact (L7)	the left side is one differential, dF
autonomous (L8)	no solving: read the phase line

which method? — the sorter

Test in this order. Stop at the first yes.

If the equation	Use	
fits $y' + p(x)\,y = q(x)$ after you divide	linear, seven lines	(L3)
factors as $y' = g(x)\,h(y)$	separable	(L4)
fits $y' + py = qy^n$, $n \neq 0, 1$	Bernoulli, $v = y^{1-n}$	(L6)
has a right side in $\frac{y}{x}$ only	homogeneous, $v = \frac{y}{x}$	(L6)
is $M\,dx + N\,dy = 0$ with $M_y = N_x$	exact, build F	(L7)
has $M_y \neq N_x$	integrating factor $\mu(x)$ or $\mu(y)$	(L7)

Trap: Some equations fit two rows. $y' + 3y = 6$ is linear and separable. Either method works.
Trap: Bernoulli with $n = 0$ or $n = 1$ is already linear. Do not substitute.

which tool? — given → find

Given → find	Do this	
an equation → order	highest derivative	(L1)
an equation → linear or not	the linear checklist	(L1)
an equation → p and q	standard form	(L1)
a y and a DE → solution or not	differentiate y, substitute	(L1)
$y' = f$ and a point → slope	evaluate f there	(L2)
f → zero-slope set	factor $f = 0$	(L2)
f with no x, y_0 → limit	nearest flat line in the drift	(L2)
linear → general solution	seven lines	(L3)
separable → general solution	separate, integrate, free y	(L4)
general solution, IC → C	plug in x_0, y_0	(L3, L4)
separable → lost solutions	roots of $h(y) = 0$	(L4)
linear IVP → interval	bad x of p and q	(L5)
nonlinear IVP → unique or not	Test 1, Test 2	(L5)
Bernoulli → solution	$v = y^{1-n}$, then L3	(L6)
$F(y/x)$ → solution	$v = y/x$, then L4	(L6)
$M\,dx + N\,dy$ → exact or not	$M_y = N_x$	(L7)
exact → solution	build F, answer $F = C$	(L7)
not exact → multiplier	one-letter ratio test	(L7)
$y' = f(y)$ → equilibria, stability	phase line	(L8)
$y' = f(y)$, y_0 → limit	follow the arrow	(L8)
words → a DE	the translation table	(L9)
a second reading → k	plug in, take ln	(L9)
tank → $Q(t)$	in − out, linear	(L9)

which way? — differentiate or integrate

Task	Direction
verify a solution	differentiate the given y
check μ	differentiate μ, compare with $p\mu$
uniqueness test	differentiate f in y (x is a constant)
linear line 6, separable step 3	integrate

Trap: To verify, never solve the DE for y'. Get y' from the given y.

integrals (add $+C$ once)

Function	Antiderivative
x^n, $n \neq -1$	$\frac{x^{n+1}}{n+1}$
y^{-n}, $n \neq 1$	$\frac{y^{1-n}}{1-n}$ *(only $n = 1$ gives ln)*
$\frac{k}{x}$	$k \ln\lvert x\rvert$ not $\ln\frac{k}{x}$
$\frac{1}{ax+b}$	$\frac{1}{a}\ln\lvert ax+b\rvert$
e^{kx}	$\frac{1}{k}e^{kx}$ divide by k
$\sin(ax)$ / $\cos(ax)$	$-\frac{\cos(ax)}{a}$ / $\frac{\sin(ax)}{a}$
$\tan x$	$\ln\lvert\sec x\rvert$
$\sec^2 x$	$\tan x$
$\frac{1}{1+y^2}$	$\tan^{-1} y$

By parts: $\int u\,dv = uv - \int v\,du$. Pick u = the part that gets simpler when you differentiate (x, x^2, $\ln x$).
Product rule (the heart of L3): $(\mu y)' = \mu' y + \mu y'$.

exponent and log rules

Rule	Use
$e^{k\ln u} = u^k$	simplify μ: $e^{2\ln x} = x^2$, $e^{-\ln x} = \frac{1}{x}$
$\frac{1}{e^{-kx}} = e^{kx}$	line 7: divide by μ
$\frac{1}{\sec x} = \cos x$	line 7 when $\mu = \sec x$
$e^{a+C} = e^C e^a$	L4: rename $e^C \to C$
e^{ab}	does not split: $e^{xy} \neq e^x e^y$
$\ln(e^{\square}) = \square$	ln and e undo each other

Lesson 1 — classify, standard form, verify

EXPLANATION

THE IDEA
Label the equation before you solve it: order, linear or not, homogeneous or not. Then put it in standard form, so that you can read p and q.

Pieces

Piece	Where to find it
$a(x)$	the coefficient of y', with all y-terms on the left
$p(x)$	the coefficient of y, after you divide by a. It keeps its sign
$q(x)$	the right side, after you divide by a

Read the pieces

Quick example	Work → answer
$xy'+2y=x^2$	$a=x$: $y'+\frac{2}{x}y=x$
$y'=3y+x$	move: $y'-3y=x$. $p=-3$
$y'+y\sin x=0$	$p=\sin x$, $q=0$: homogeneous
$y'=xy^2$	y^2: not linear. No p, no q

Linear or not?

Still linear	Breaks linearity
x^2y', e^xy, $\sin x$	y^2, yy', $(y')^3$
a missing term (coefficient 0)	$\sin y$, e^y

Trap: p includes its sign. $y'-3y$ has $p=-3$.
Trap: Divide every term by a, the right side too.
Trap: Homogeneous means q is the zero function. Do not plug in $x=0$.
Trap: To verify, never solve the DE for y'. Get y' from the given y.

WORK

Steps — classify

1. ORDER
The highest derivative present. Exponents do not matter: $(y'')^3$ is order 2.

↓

2. LINEAR?
Every $y, y', y'', \ldots$ appears alone, to the first power, with a coefficient in x only. Never x-stuff breaks it.

↓

3. HOMOGENEOUS? (2.1 SENSE)
Move all y-terms to the left. If the right side is the zero function, it is homogeneous.

Steps — standard form

1. MOVE
All y-terms to the left: $\boxed{a}y'+by=g$

↓

2. DIVIDE EVERY TERM BY a
$y'+\boxed{\frac{b}{a}}y=\boxed{\frac{g}{a}}$ *(the coefficient of y' must be 1)*

↓

3. READ
$p=\frac{b}{a}$ $\quad q=\frac{g}{a}$ *(write both down)*

Steps — verify a solution

1. DIFFERENTIATE
Find y' (and y'' if needed) from the given y, with calculus.

↓

2. SUBSTITUTE
Put y, y', … into the left side only.

↓

3. SIMPLIFY AND COMPARE
Simplify the left side. If it equals the right side for every x, y is a solution.

Lesson 2 — direction fields

EXPLANATION

THE IDEA
$y' = f(x,y)$ gives the slope at every point. The field is a picture of those slopes. You never solve anything.

Pieces

Piece	Where to find it
$f(x,y)$	the right side, after you get y' alone
(x_0,y_0)	the point you are asked about

Sign of the slope

Slope	The solution
$y' > 0$	rises as x increases
$y' < 0$	falls
$y' = 0$	is flat

Does f have x?

If f	Then
has no x: $y' = f(y)$	every column is the same. Zero-slope lines are horizontal (equilibria). Solutions approach them and never cross
has x	the zero-slope set is a curve, not a wall. No equilibria

Trap: Never divide $f = 0$ by y. You lose the root $y = 0$.
Trap: The limit only works when f has no x.

WORK

Steps — slope at a point

1. TEMPLATE
slope $= \boxed{f}\left(\boxed{x_0}, \boxed{y_0}\right)$

↓

2. READ THE SIGN
$+$ rises, $-$ falls, 0 flat.

Steps — zero-slope set

1. SET
$\boxed{f(x,y)} = 0$

↓

2. FACTOR
Pull out every common factor. Do not divide.

↓

3. EACH FACTOR $= 0$
Each factor gives one line or curve of the set.

Steps — limit as $x \to \infty$ (f has no x)

1. FLAT LINES
Solve $\boxed{f(y)} = 0$. List the roots.

↓

2. DRIFT
Sign of $\boxed{f}\left(\boxed{y_0}\right)$: $+$ moves up, $-$ moves down, 0 stays.

↓

3. LIMIT
The nearest flat line in the direction of drift. If there is none in that direction: $\pm\infty$.

Lesson 3 — first order linear (seven lines)

EXPLANATION

THE IDEA

Multiply by μ so that the left side becomes one product-rule derivative, $(\mu y)'$. Then one integral finishes the job.

Pieces

Piece	Where to find it
p	the coefficient of y in standard form, sign included
q	the right side in standard form
μ (mu)	you make it: $e^{\int p\,dx}$. No $+C$
x_0, y_0	the initial condition. Use it only at the end

Line 2 — common μ

If p is	then μ is
k (a constant)	e^{kx}
$\frac{k}{x}$	$e^{k\ln x} = x^k$
$-\frac{1}{x}$	$e^{-\ln x} = \frac{1}{x}$
$2x$	e^{x^2}
$\tan x$	$e^{\ln\lvert\sec x\rvert} = \sec x$
$-\tan x$	$e^{\ln\lvert\cos x\rvert} = \cos x$

Equivalences

This	is the same as
μ'	$p\mu$, always (that is line 3)
$\mu p\,y$	a line 4 term
μy	the line 6 left side
$\frac{1}{e^{-kx}}$ / $\frac{1}{\sec x}$	e^{kx} / $\cos x$

Trap: $\int \frac{k}{x}\,dx = k\ln x$, not $\ln\frac{k}{x}$.
Trap: $\int e^{kx}dx = \frac{1}{k}e^{kx}$. Divide by k.
Trap: $\int \tan x\,dx = \ln\lvert\sec x\rvert$.
Trap: Line 7: divide C by μ too.

WORK

The seven lines

LINE 1 — STANDARD FORM, READ THE PIECES

$y' + \boxed{p}y = \boxed{q}$ Write $p = \ldots$ and $q = \ldots$

↓

LINE 2 — BUILD μ

$\boxed{\mu} = e^{\int \boxed{p}dx}$ Simplify. $(e^{k\ln u} = u^k)$ No $+C$.

↓

LINE 3 — CHECK μ

Differentiate: $\boxed{\mu}' = \ldots$ Separately: $\boxed{p}\boxed{\mu} = \ldots$
They must be equal. If not, fix line 2.

↓

LINE 4 — MULTIPLY EVERY TERM

$\boxed{\mu}y' + \boxed{\mu}\boxed{p}y = \boxed{\mu}\boxed{q}$
(three multiplications, written out)

↓

LINE 5 — COLLAPSE THE LEFT SIDE

$\left(\boxed{\mu}y\right)' = \boxed{\mu}\boxed{q}$
$((\mu y)' = \mu' y + \mu y' = \mu p\,y + \mu y'$: the whole left side of line 4)

↓

LINE 6 — INTEGRATE BOTH SIDES

$\boxed{\mu}y = \int \boxed{\mu}\boxed{q}dx + C$
(left side is μ times y: not μ', not μp)

↓

LINE 7 — DIVIDE BY μ

$y = \frac{1}{\boxed{\mu}}\left(\int \boxed{\mu}\boxed{q}dx + C\right)$
(divide the whole parenthesis: C becomes C/μ)

↓

IC (IF GIVEN)

Plug in x_0 and y_0. Solve for C. Rewrite the answer with the number.

Lesson 4 — separable

EXPLANATION

THE IDEA

Factor the right side into an x-part times a y-part. Move all y to the dy side and all x to the dx side. Integrate each side.

Pieces

Piece	Where to find it
$g(x)$	the factor with only x (a constant counts)
$h(y)$	the factor with only y
x_0, y_0	the initial condition

Read the pieces

Quick example	Work → answer
$y' = xy$	$g = x$, $h = y$
$y' = \frac{x}{y}$	$g = x$, $h = \frac{1}{y}$
$y' = e^{x+y}$	$= e^x e^y$: $g = e^x$, $h = e^y$
$y' = 2y$	$g = 2$, $h = y$
$y' = x + y$	will not factor: not separable

Step 4 — free y

Wrapped	Undo with	Watch for
$\ln\lvert y\rvert = \ldots$	$e^{(\cdot)}$ of the whole side	$e^{a+C} = e^C e^a$. Rename $e^C \to C$
$y^2 = \ldots$	square root	$y = \pm\sqrt{\ldots}$. Keep $\pm$ unless an IC picks one
$\frac{1}{y} = \ldots$	reciprocal	do it first, before anything else
$\tan^{-1} y = \ldots$	tan	$y = \tan(\ldots + C)$, C stays inside

Trap: $\int y^{-n}\,dy = \frac{y^{1-n}}{1-n}$ for $n \neq 1$. Only $n = 1$ gives ln.
Trap: Keep the dy and the dx when you separate.

WORK

Steps

1. FACTOR

$\frac{dy}{dx} = \boxed{g(x)}\,\boxed{h(y)}$ If it will not factor, it is not separable.

↓

2. SEPARATE

$\frac{dy}{\boxed{h(y)}} = \boxed{g(x)}\,dx$

↓

3. INTEGRATE

$\int \frac{dy}{\boxed{h(y)}} = \int \boxed{g(x)}\,dx + C$ *(one $+C$ only)*

↓

4. FREE y

Apply the inverse of whatever wraps y. Use the table on the left.

↓

5. IC (IF GIVEN)

Plug in x_0, y_0. Solve for C.

↓

6. LOST SOLUTIONS

Solve $\boxed{h(y)} = 0$. Each root $y = k$ is a constant solution. Step 2 divided by it. Mention them if asked for all solutions.

Lesson 5 — existence and uniqueness

THE IDEA
Before you solve, ask: does a solution through the point exist, and is it the only one? Linear and nonlinear equations use different tests.

Linear: the interval

EXPLANATION

Pieces

Piece	Where to find it
p, q	standard form $y' + py = q$
x_0	the IC $y(x_0) = y_0$. y_0 does not matter

Bad x

Undefined	Bad x
division by 0	where the bottom $= 0$
$\ln(\dots)$	where $\dots \le 0$
even root $\sqrt{\dots}$	where $\dots < 0$
$\tan x$	$x = \frac{\pi}{2} + n\pi$

Read the pieces

Quick example	Work → answer
$p = \frac{1}{x-3}$, $x_0 = 1$	bad: 3. $(-\infty, 3)$
$p = \ln x$, $x_0 = 2$	bad: $x \le 0$. $(0, \infty)$

Trap: Divide by $a(x)$ first. Its zeros become bad x.

WORK

1. STANDARD FORM
$y' + \boxed{p}\,y = \boxed{q}$

↓

2. BAD x OF p
List every x where p is undefined.

↓

3. BAD x OF q
List every x where q is undefined.

↓

4. NUMBER LINE
Mark all bad x and x_0.

↓

5. INTERVAL
(nearest bad x below x_0, nearest bad x above x_0). Open. No bad x on a side: $\pm\infty$.

Nonlinear: Test 1 and Test 2

EXPLANATION

Pieces

Piece	Where to find it
$f(x,y)$	the right side of $y' = f(x,y)$
$\frac{\partial f}{\partial y}$	you make it: differentiate f in y, x held constant
(x_0, y_0)	the IC. Both numbers matter

Verdict

Result	Verdict
both pass	a unique solution
Test 1 yes, Test 2 no	exists, uniqueness not guaranteed
Test 1 fails	the theorem says nothing

Read the pieces

Quick example	Work → answer
$f = y^{1/3}$, $y_0 = 0$	$\frac{1}{3y^{2/3}}$ breaks at 0: exists, not guaranteed unique
$f = y^{1/3}$, $y_0 = 1$	both pass: unique

Trap: The tests are at the point. Changing y_0 can change the verdict.
Trap: A point on the edge of the domain fails Test 1: $\sqrt{y}$ at $y_0 = 0$ is not defined below.

WORK

1. NORMAL FORM
$y' = \boxed{f(x,y)}$

↓

2. TEST 1 — EXISTENCE
$\boxed{f}\left(\boxed{x_0}, \boxed{y_0}\right)$ is a real number, and f is defined on an open rectangle around the point.

↓

3. PARTIAL
$\boxed{\frac{\partial f}{\partial y}} = \dots$ *(x is a constant)*

↓

4. TEST 2 — UNIQUENESS
$\boxed{\frac{\partial f}{\partial y}}\left(\boxed{x_0}, \boxed{y_0}\right)$ is a real number.

↓

5. VERDICT
Use the verdict table on the left.

Lesson 6 — transformations

THE IDEA
Substitute a new variable so that the equation becomes a type you can already solve. Bernoulli → linear (L3). Homogeneous → separable (L4).

Bernoulli: $y' + p\,y = q\,y^n$

EXPLANATION

Pieces

Piece	Where to find it
p	the coefficient of y on the left
q	the coefficient of y^n on the right
n	the power on the right. $n \neq 0, 1$
new p, new q	$(1-n)\,p$ and $(1-n)\,q$ (step 5)

Read the pieces

Quick example	Work → answer
$y' + y = xy^3$	1, x, 3: $v = y^{-2}$. new -2, new $-2x$
$y' - \frac{y}{x} = y^2$	$-\frac{1}{x}$, 1, 2: $v = y^{-1}$. new $\frac{1}{x}$, new -1

Trap: Keep the $(1-n)$. It is often negative.
Trap: Divide every term by y^n, the y' term too.
Trap: Forget the $(1-n)$ in step 5 and every later line is wrong.

WORK

1. MATCH, READ THE PIECES
$y' + \boxed{p}\,y = \boxed{q}\,y^{\boxed{n}}$

↓

2. CHOOSE v
$v = y^{1-\boxed{n}} \qquad v' = (1-\boxed{n})\,y^{-\boxed{n}}\,y'$

↓

3. DIVIDE EVERY TERM BY y^n
$y^{-\boxed{n}}y' + \boxed{p}\,y^{1-\boxed{n}} = \boxed{q}$

↓

4. REPLACE
$y^{-n}y' \to \dfrac{v'}{1-n}, \quad y^{1-n} \to v: \qquad \dfrac{v'}{1-\boxed{n}} + \boxed{p}\,v = \boxed{q}$

↓

5. MULTIPLY BY $(1-n)$, READ NEW PIECES
$v' + \underbrace{(1-\boxed{n})\boxed{p}}_{\text{new } p}\,v = \underbrace{(1-\boxed{n})\boxed{q}}_{\text{new } q}$

Step 6 — the seven lines on v

LINE 1 — STANDARD FORM IN v
$v' + \boxed{\text{new } p}\,v = \boxed{\text{new } q}$ *(from step 5)*

↓

LINE 2 — BUILD μ
$\boxed{\mu} = e^{\int \boxed{\text{new } p}\,dx}$ Simplify. No $+C$.

↓

LINE 3 — CHECK μ
Differentiate: $\boxed{\mu}' = \ldots$ Separately: $\boxed{\text{new } p}\,\boxed{\mu} = \ldots$
They must be equal.

↓

LINE 4 — $\boxed{\mu}$LTIPLY EVERY TERM
$\boxed{\mu}\,v' + \boxed{\mu}\,\boxed{\text{new } p}\,v = \boxed{\mu}\,\boxed{\text{new } q}$

↓

LINE 5 — COLLAPSE THE LEFT SIDE
$\left(\boxed{\mu}\,v\right)' = \boxed{\mu}\,\boxed{\text{new } q}$

↓

LINE 6 — INTEGRATE BOTH SIDES
$\boxed{\mu}\,v = \int \boxed{\mu}\,\boxed{\text{new } q}\,dx + C$ *(μ times v, not y)*

↓

LINE 7 — DIVIDE BY μ
$v = \dfrac{1}{\boxed{\mu}}\left(\int \boxed{\mu}\,\boxed{\text{new } q}\,dx + C\right)$ *(C becomes C/μ)*

↓

7. BACK-SUBSTITUTE
$y = v^{1/(1-\boxed{n})}$ *(raise the whole line 7 answer, C included)*

↓

8. IC (IF GIVEN)
Plug x_0, y_0 into the y answer. Solve for C.

Homogeneous: $y' = F(y/x)$

EXPLANATION

Pieces

Piece	Where to find it
v	$v = \frac{y}{x}$, so $y = vx$
$F(v)$	the right side, rewritten in $\frac{y}{x}$ only, with $\frac{y}{x} \to v$

Get F: divide the top and the bottom by the highest power of x.

Quick example	Work → answer
$y' = \dfrac{x+y}{x}$	$1 + \frac{y}{x}$: $F = 1 + v$
$y' = \dfrac{x^2+y^2}{xy}$	$\div x^2$: $F = \frac{1+v^2}{v}$
$y' = \dfrac{y}{x} + \dfrac{x}{y}$	$F = v + \frac{1}{v}$

Trap: y' is not v'. $y' = v + xv'$.

Two meanings of "homogeneous"

Where	Means
Lesson 1 (Trench 2.1)	the right side of $y' + py = q$ is the zero function: $q = 0$
Lesson 6	the right side is a function of $\frac{y}{x}$ only: $y' = F(y/x)$

WORK

1. REWRITE

$y' = \boxed{F}\left(\frac{y}{x}\right)$

2. SUBSTITUTE

$y = \boxed{v}x \qquad y' = \boxed{v} + x\boxed{v}'$ *(product rule)*

3. PLACE AND CANCEL

$\boxed{v} + xv' = \boxed{F(v)} \;\Rightarrow\; xv' = \boxed{F(v)} - \boxed{v}$

4. SEPARATE (LESSON 4)

$\dfrac{dv}{\boxed{F(v)} - \boxed{v}} = \dfrac{dx}{x}$ $\quad (g(x) = \frac{1}{x},\ \ h(v) = F(v) - v)$

5. INTEGRATE

Both sides. One $+C$. $\int \frac{dx}{x} = \ln|x|$.

6. BACK-SUBSTITUTE

Replace v with $\frac{y}{x}$. Free y if you can.

Lesson 7 — exact equations

THE IDEA
Find one function $F(x,y)$ whose partial derivatives are M and N. Then the whole equation says $dF = 0$, so the answer is $F = C$.

Exact: build F

EXPLANATION

Pieces

Piece	Where to find it
$M(x,y)$	whatever multiplies dx
$N(x,y)$	whatever multiplies dy
$g(y)$	the "constant" of the x-integral: an unknown function of y
$F(x,y)$	you build it: $F_x = M$, $F_y = N$

Read the pieces

Quick example	Work → answer
$(2xy+1)\,dx + (x^2+3y^2)\,dy = 0$	$M = 2xy+1$, $N = x^2+3y^2$
$y' = -\dfrac{x}{y}$	$y\,dy = -x\,dx \to x\,dx + y\,dy = 0$: $M = x$, $N = y$

Equivalences

This	is the same as
M	F_x
N	F_y
the test $M_y = N_x$	mixed partials agree: $F_{xy} = F_{yx}$

Trap: Integrate M in x: y is a constant. The "$+C$" is $g(y)$, not a number.
Trap: In step 5, every x term must cancel. If one is left, a step above is wrong.
Trap: The answer is $F = C$, not $F = 0$. Implicit is fine.

WORK

1. FORM, READ THE PIECES
$\boxed{M}\,dx + \boxed{N}\,dy = 0$

↓

2. TEST
$\dfrac{\partial \boxed{M}}{\partial y} \stackrel{?}{=} \dfrac{\partial \boxed{N}}{\partial x}$ Equal → exact. Not equal → stop, or find μ (below).

↓

3. INTEGRATE M IN x
$\boxed{F} = \displaystyle\int \boxed{M}\,dx + \boxed{g(y)}$ *(y held constant)*

↓

4. DIFFERENTIATE F IN y
$\dfrac{\partial \boxed{F}}{\partial y} = \ldots + \boxed{g'(y)}$

↓

5. SET EQUAL TO N, SOLVE FOR g'
$\ldots + \boxed{g'(y)} = \boxed{N}$ *(all x terms cancel)*

↓

6. INTEGRATE g'
$\boxed{g(y)} = \displaystyle\int \boxed{g'(y)}\,dy$ Put it back into step 3.

↓

7. ANSWER
$\boxed{F(x,y)} = C$ IC given: plug in x_0, y_0 for C.

Not exact: find an integrating factor μ

EXPLANATION

Pieces

Piece	Where to find it
M, N	as above
M_y, N_x	from the failed test
μ	you make it: $\mu(y)$ or $\mu(x)$, one letter only

Definition: μ is an integrating factor if $\mu M\,dx + \mu N\,dy = 0$ is exact.

Which μ?

If this has only one letter	then
$\dfrac{N_x - M_y}{M}$ has only y	$\mu(y) = e^{\int (\text{that})\,dy}$
$\dfrac{M_y - N_x}{N}$ has only x	$\mu(x) = e^{\int (\text{that})\,dx}$

Trap: $\mu(y)$ has no x, so $\frac{\partial}{\partial x}(\mu N) = \mu N_x$. Only the y-side needs the product rule.
Trap: No $+C$ in μ.

WORK

1. MULTIPLY
$\boxed{\mu}\boxed{M}\,dx + \boxed{\mu}\boxed{N}\,dy = 0$

↓

2. REQUIRE EXACT
$\dfrac{\partial(\boxed{\mu}\boxed{M})}{\partial y} = \dfrac{\partial(\boxed{\mu}\boxed{N})}{\partial x}$

↓

3. EXPAND, FOR $\mu = \mu(y)$
$\boxed{\mu}'\boxed{M} + \boxed{\mu}\boxed{M_y} = \boxed{\mu}\boxed{N_x}$ *(product rule on the left only)*

↓

4. SEPARATE (LESSON 4)
$\dfrac{\boxed{\mu}'}{\boxed{\mu}} = \dfrac{\boxed{N_x} - \boxed{M_y}}{\boxed{M}}$ Must have only y.
$\boxed{\mu} = e^{\int (\ldots)\,dy}$

↓

5. NEW PIECES, REDO
new $M = \mu M$, new $N = \mu N$. Do exact steps 2–7.

Lesson 8 — autonomous equations

EXPLANATION

THE IDEA

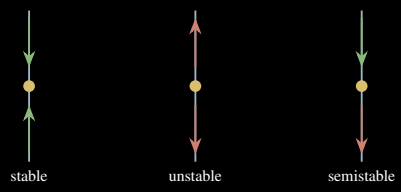
$\frac{dy}{dt} = f(y)$ has no t on the right. This is the Lesson 2 "no x" case. You read the long-term behavior off a phase line, without solving.

Pieces

Piece	Where to find it
$f(y)$	the right side. No t in it
equilibria	the roots of $f(y) = 0$: constant solutions
y_0	the IC $y(0) = y_0$

Classify an equilibrium

Arrows	Type
both point toward it	stable (asymptotically stable, sink)
both point away	unstable (source)
one toward, one away	semistable



Sign → arrow

Sign of f	Arrow
$f > 0$	up (y increases)
$f < 0$	down (y decreases)

Trap: Never divide $f = 0$ by y. You lose the equilibrium $y = 0$.
Trap: Test one y inside each region, not at an equilibrium.
Trap: A solution never crosses an equilibrium.

WORK

1. EQUILIBRIA
Solve $\boxed{f(y)} = 0$. Factor, set each factor to 0.
List the equilibria in order on a vertical line.

↓

2. SIGN CHART
In each region between equilibria, pick one y. Find the sign of $\boxed{f}(y)$.

↓

3. PHASE LINE
Draw an arrow in each region: $+$ up, $-$ down.

↓

4. CLASSIFY
Look at the arrows on both sides of each equilibrium. Use the table on the left.

↓

5. LIMIT (NO SOLVING)
Find the region of y_0. Follow its arrow to the nearest equilibrium in that direction. That is $\lim_{t \to \infty} y(t)$.
No equilibrium in that direction: $\pm\infty$. y_0 on an equilibrium: it stays.

Lesson 9 — applications

THE IDEA
Every application is the same chain: words → a DE → separable or linear → solve → fit the constants. Usually two constants: C from the IC, and a rate constant k from a second data point.

The recipe

EXPLANATION

Words → math

Words	Math
"rate of change of Q"	$\frac{dQ}{dt}$
"proportional to"	$k\cdot$
"at a rate equal to 25% of"	$0.25\cdot$
"decreases", "decays", "loses"	a minus sign
"initially"	$t=0$
"6 months ago"	pick $t=0$ as now or as then. Stay consistent
"of the remaining"	of the current amount Q

Trap: Write the units of t down once. Every rate uses them.

WORK

1. TRANSLATE
Words → a DE. Use the table on the left.

↓

2. CLASSIFY
Separable (L4) or linear (L3)?

↓

3. SOLVE
General solution with C (and k).

↓

4. C FROM THE IC
Plug in $t=0$.

↓

5. k FROM A SECOND DATA POINT
Plug in (t_1, value_1). Solve for k: usually a ln.

↓

6. ANSWER THE QUESTION
A value at a time: plug in t. A time: solve for t with ln.

Newton's law of cooling

EXPLANATION

Pieces

Piece	Where to find it
T_m	the surrounding (room) temperature
T_0	the temperature at $t=0$
k	the rate constant, $k>0$. Unknown
(t_1, T_1)	the second reading

Trap: Subtract T_m first. The ratio is $\frac{T_1-T_m}{T_0-T_m}$, not $\frac{T_1}{T_0}$.
Trap: $T \to T_m$ as $t \to \infty$, from above or below.

WORK

1. DE
$\frac{dT}{dt} = -\boxed{k}\left(T - \boxed{T_m}\right)$ *(separable and linear)*

↓

2. SHAPE
$T = \boxed{T_m} + \left(\boxed{T_0} - \boxed{T_m}\right)e^{-\boxed{k}t}$

↓

3. k FROM THE READING
$e^{-\boxed{k}\boxed{t_1}} = \frac{\boxed{T_1}-\boxed{T_m}}{\boxed{T_0}-\boxed{T_m}} \Rightarrow \boxed{k} = -\frac{1}{\boxed{t_1}}\ln(\text{that ratio})$

↓

4. ANSWER
T at a time: plug in t. Time to reach T: $t = -\frac{1}{k}\ln\frac{T-T_m}{T_0-T_m}$.

Exponential growth and decay

EXPLANATION

Pieces

Piece	Where to find it
M_0	the amount at $t=0$
k	the rate. $k>0$ grows, $k<0$ decays
data	a second amount (t_1, M_1), or a half-life T

Which k?

Words	k
"decays at 25% of the remaining mass per hour"	$k=-0.25$ directly
"loses 25% each hour" (the amount after 1 h)	$e^k = 0.75$, $k = \ln 0.75$
amount M_1 at t_1	$k = \frac{\ln(M_1/M_0)}{t_1}$
half-life T	$\frac{1}{2} = e^{kT}$, $k = -\frac{\ln 2}{T}$

WORK

1. DE
$\frac{dM}{dt} = \boxed{k}M$ *(separable)*

↓

2. SHAPE
$M = \boxed{M_0}e^{\boxed{k}t}$

↓

3. k FROM THE DATA
Use the "Which k?" table on the left.

↓

4. ANSWER
M at a time: plug in t. Time to reach M: $t = \frac{\ln(M/M_0)}{k}$.

Mixing (tank)

EXPLANATION

Pieces

Piece	Where to find it
$Q(t)$	the amount of salt. The unknown
c_{in}	the concentration coming in. Pure water: 0
r_{in}, r_{out}	the flow in and the flow out (volume per time)
V_0, Q_0	the starting volume and amount
Δ	$r_{\text{in}} - r_{\text{out}}$, the change in volume per time

Lesson 3 pieces

Piece	Value
$p(t)$	$\dfrac{r_{\text{out}}}{V_0 + \Delta t}$
$q(t)$	$c_{\text{in}}\, r_{\text{in}}$
μ, $\Delta = 0$	$e^{(r_{\text{out}}/V_0)t}$
μ, $\Delta \neq 0$	$(V_0 + \Delta t)^{r_{\text{out}}/\Delta}$ *(a power of $V(t)$)*

Trap: Rate out uses the tank concentration $\frac{Q}{V}$, not c_{in}.
Trap: Concentration $= \frac{Q}{V}$. If asked for concentration, divide at the end.
Trap: Units: (amount/volume)(volume/time) = amount/time.

WORK

1. VOLUME
$V(t) = \boxed{V_0} + (\boxed{r_{\text{in}}} - \boxed{r_{\text{out}}})t = \boxed{V_0} + \boxed{\Delta}t$

2. DE: IN − OUT
$$\frac{dQ}{dt} = \boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}} - \frac{Q}{V(t)}\,\boxed{r_{\text{out}}}$$

3. STANDARD FORM (LINE 1)
$Q' + \dfrac{\boxed{r_{\text{out}}}}{V(t)}\,Q = \boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}}$ Read p, q (left table).

4. BUILD AND CHECK μ (LINES 2–3)
$\mu = e^{\int p\,dt}$. Use the left table. Check $\mu' = p\mu$.

5. MULTIPLY, COLLAPSE (LINES 4–5)
$(\mu Q)' = \mu\,\boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}}$

6. INTEGRATE (LINE 6)
$$\mu Q = \int \mu\,\boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}}\,dt + C$$

7. DIVIDE (LINE 7)
$$Q = \frac{1}{\mu}\left(\int \mu\, c_{\text{in}} r_{\text{in}}\,dt + C\right)$$

8. C FROM Q_0
Plug in $t = 0$, $Q = \boxed{Q_0}$.

Growth proportional to a power

EXPLANATION

Pieces

Piece	Where to find it
a	the power: "proportional to $\sqrt{W}$" gives $a = \frac{1}{2}$
k	the unknown constant
two data points	$(0, W_0)$ gives C, (t_1, W_1) gives k

Trap: $\int W^{-1/2}\,dW = 2W^{1/2}$. Divide by $\frac{1}{2}$.
Trap: Find C first from $t = 0$. Then k.

WORK

1. DE
$$\frac{dW}{dt} = \boxed{k}\,W^{\boxed{a}} \quad \textit{(separable)}$$

2. SEPARATE
$W^{-\boxed{a}}\,dW = \boxed{k}\,dt$

3. INTEGRATE
$$\frac{W^{1-\boxed{a}}}{1-\boxed{a}} = \boxed{k}t + C \quad (a = \tfrac{1}{2}\text{: } 2\sqrt{W} = kt + C)$$

4. C, THEN k
Plug in $(0, W_0)$ for C. Then (t_1, W_1) for k.

5. FREE W
$a = \frac{1}{2}$: $W = \left(\dfrac{kt + C}{2}\right)^2$